For teachers only. Do not give this part to students.

Research conditions, distribution and timing

This group uses the worksheet without operating the VL or receiving a lesson beforehand. Both groups have 15–20 minutes in total. Use the same T for both groups in each session, including self-study, practice and checking.

Choose a common T = _____ minutes (15–20) for both groups. Select it for the class reading pace and record actual start/end times. Checking is included; do not add a lesson beforehand.

Stage	T	T=15	T=20
A: notes, examples and experiment figures	30%	4.5 min	6 min
B1–B2: guided practice	35%	5.25 min	7 min
B3–B4: independent application	25%	3.75 min	5 min
C: check and correct	10%	1.5 min	2 min

Distribution and teacher support

1. Give out A and B at the start. Keep A available. Explain only reading, answering and timing, not digestion concepts.
2. In B1–B2, clarify wording/procedure and point to A, without giving answers or deciding for students. Log extra clarification and use the common support rules for both groups.
3. B3–B4 are independent, with A available. Address procedure/access needs only, without answer feedback.
4. Save the first B answers before distributing C at checking time. Correct in another colour. Give brief consistent feedback using C, keeping first and corrected answers separate.
5. Give students only A–C. D and the complete pack are for teacher printing. Verify comparable actual total time from session records, not just a screen label.

Draft worksheet feedback guide. These marks are not equivalent to VL activity marks, SPSAI or CKT. Use separate common tests to assess research outcomes.

This material follows the specified self-study comparison condition; its activity process differs from IBL.

For teachers only. Do not give this part to students.

Full answer key and teaching limits

B1.1 • Describe only the oil–water test tube before any digestive juice is added. What layers and boundary can you see?

Oil is above the water, with a clear boundary. Keep the visible description separate from a chemical explanation.

B1.2 • Judge whether this prediction is testable and its reason is correct. Explain briefly.

The prediction can be tested, but the reason is wrong. Y is lipase (脂肪酶), which speeds up hydrolysis. Bile (膽汁) causes emulsification (乳化作用). Testable does not mean already proved.

B1.3 • The main comparison changes the combination of digestive juices. Classify the factors as independent, dependent or controlled variables.

Independent: combination of juices. Dependent: appearance. Controlled: starting oil/water volume, temperature, time, total added volume and shaking method.

B1.4 • Choose all the conditions needed for a fair comparison.

Choose equal time, equal starting and total added volumes, and equal shaking; do not choose extra total liquid for X + Y.

B2.1 • To compare adding digestive juice with adding only water, which of the four sets is the baseline? Why keep the total volume added the same?

The 2 mL water set is the baseline for this question. Equal added volume avoids a difference in total volume or dilution.

B2.2 • The four basic setups are already specified. Use the recipes in A to complete each label and added volume; no new apparatus design is needed.

In order: water control, 2 mL water; X, 2 mL; Y, 2 mL; X + Y, 1 mL each.

B2.3 • Someone adds 2 mL water to the control but 2 mL X and 2 mL Y to the combined set, with other conditions equal. Is this fair? Identify the problem, fix the two volumes and explain why.

Unfair: 2 mL versus 4 mL. Use 1 mL of each juice, keeping the total at 2 mL to avoid a volume/dilution difference.

Appearance is a teaching clue: clarity alone does not prove glycerol (甘油) and fatty acids (脂肪酸) formed or measure speed. X + Y uses 1 mL each, while a single-juice set uses 2 mL. The different individual amounts cannot prove a measured combined effect or faster rate.

For teachers only. Do not give this part to students.

Full answer key and teaching limits

B3.1 • Use Figure 1 on A2 to record the four appearances.

Water: separated layers; X: cloudy; Y: thinner oil layer, still separated; X + Y: clear.

B3.2 • X: choose the appearance, then explain it using emulsification (乳化作用).

Cloudy emulsion (乳状液). X disperses oil droplets by emulsification (乳化作用), a form of physical digestion (物理消化), without changing the lipid molecules.

B3.3 • Total surface area after droplets become smaller: choose the answer and explain how this helps Y act.

Increases. For the same total amount of lipid, small droplets provide more surface for lipase (脂肪酶) to act on. This is not a measurement of speed.

B4.1 • Limits of appearance evidence: choose the best statement, then name another kind of evidence needed.

Choose the statement that appearance gives clues but does not prove glycerol (甘油) and fatty acids (脂肪酸) formed. Other chemical product tests and repeated tests are needed.

B4.2 • To study boiling one digestive juice, which set should be the baseline? What must happen to the boiled liquid before comparison?

The unboiled X + Y set, 1 mL each; cool the boiled juice to the common 37°C and keep the other conditions equal. This control contains X and Y.

B4.3 • Use Figure 2 on A2 to record the heated sets and compare them with unboiled X + Y.

Boiled X + unboiled Y: clear; unboiled X + boiled Y: cloudy.

B4.4 • Juice that loses its action after heating: choose the answer and explain why cooling does not restore its action.

Y. Lipase (脂肪酶) is a protein; boiling causes denaturation and loss of function. Cooling does not restore it in this model.

B4.5 • Use at least one main-set comparison and one heating comparison to explain emulsification (乳化作用), enzyme action and denaturation. Name the sets. Do not claim a measured combined effect or increase in speed.

For example, compare water with X for emulsification (乳化作用), and unboiled X + Y with boiled Y for enzyme loss. Explain bile (膽汁), lipase (脂肪酶) hydrolysis and heat denaturation. Appearance and different individual juice amounts limit measured claims. Other sound specific comparisons are accepted.

Appearance is a teaching clue: clarity alone does not prove glycerol (甘油) and fatty acids (脂肪酸) formed or measure speed. X + Y uses 1 mL each, while a single-juice set uses 2 mL. The different individual amounts cannot prove a measured combined effect or faster rate.

For teachers only. Do not give this part to students.

Question-by-question mapping

Original VL item	Treatment	Revised item	Shared goal	Reason
Your first observation： Describe only what you can see in the oil + water test tube (do not yet guess what X and Y are or what the results will be).	Retained / equivalent rewrite	B1.1 Describe only the oil–water test tube before any digestive juice is added. What layers and boundary can you see?	Observation: separate what is seen from an explanation.	Keep the common content and skills, practised through self-study or static data.
Complete your prediction / If I add _____to the oil and water, the mixture in the test tube will _____. / My reason (required; there is no single correct answer)	Rewritten	B1.2 Judge whether this prediction is testable and its reason is correct. Explain briefly.	Inquiry design: judge whether a prediction is testable and whether its reason is sound.	Teach the method with an example, then evaluate another prediction and reason.
02 • Fair test design： Independent variable (the factor changed on purpose) / Dependent variable (the result you measure or observe) / Controlled variables (the factors kept the same)	Retained / equivalent rewrite	B1.3 The main comparison changes the combination of digestive juices. Classify the factors as independent, dependent or controlled variables.	Classification: identify the three kinds of variables.	Keep the common content and skills, practised through self-study or static data.
Which conditions does this fair test need?	Retained / equivalent rewrite	B1.4 Choose all the conditions needed for a fair comparison.	Inquiry design and experimental reasoning: identify sound fixed conditions.	Keep the common content and skills, practised through self-study or static data.
Does this experiment need a control group? If so, how would you design the setups to compare?	Rewritten	B2.1 To compare adding digestive juice with adding only water, which of the four sets is the baseline? Why keep the total volume added the same?	Inquiry design: explain the purpose of a control.	Keep the common content and skills, practised through self-study or static data.

For teachers only. Do not give this part to students.

Question-by-question mapping

Original VL item	Treatment	Revised item	Shared goal	Reason
Written design (can replace a drawing): describe the four setups, how much liquid you add and which conditions you keep the same. / Draw and label your setup / Use your experimental variables to help you draw and label your setup.	Rewritten	B2.2 The four basic setups are already specified. Use the recipes in A to complete each label and added volume; no new apparatus design is needed.	Experimental reasoning and communication: read and label setups and volumes.	Label and evaluate a given procedure while practising apparatus and fair-comparison skills.
Written design (can replace a drawing): describe the four setups, how much liquid you add and which conditions you keep the same. / Does this experiment need a control group? If so, how would you design the setups to compare?	Rewritten	B2.3 Someone adds 2 mL water to the control but 2 mL X and 2 mL Y to the combined set, with other conditions equal. Is this fair? Identify the problem, fix the two volumes and explain why.	Inquiry design and experimental reasoning: evaluate and correct an unfair procedure.	Keep the common content and skills, practised through self-study or static data.
Select the appearance based on what you observe:	Static data	B3.1 Use Figure 1 on A2 to record the four appearances.	Observation and communication: distinguish the four appearances.	Keep the common content and skills, practised through self-study or static data.
1. What does digestive juice X do?：Digestive juice X mainly causes lipids to form _____.	Retained / equivalent rewrite	B3.2 X: choose the appearance, then explain it using emulsification (乳化作用).	Inference: link an appearance with emulsification (乳化作用).	Keep the common content and skills, practised through self-study or static data.
2. How does digestive juice X help digestive juice Y act?：Digestive juice X first breaks up large oil droplets into small droplets, causing their total surface area to _____, which helps digestive juice Y act.	Retained / equivalent rewrite	B3.3 Total surface area after droplets become smaller: choose the answer and explain how this helps Y act.	Inference: explain surface area without inventing speed data.	Keep the common content and skills, practised through self-study or static data.

For teachers only. Do not give this part to students.

Question-by-question mapping

Original VL item	Treatment	Revised item	Shared goal	Reason
3. What can this experiment tell us?：We only observed the oil layer and whether the mixture was cloudy or clear. We did not test for digestion products. Which statement fits best?_____	Retained / equivalent rewrite	B4.1 Limits of appearance evidence: choose the best statement, then name another kind of evidence needed.	Inference and communication: distinguish clues, direct evidence and limits.	Keep the common content and skills, practised through self-study or static data.
Does this experiment need a control group? If so, how would you design the setups to compare? / Which digestive juice is affected by high temperature?	Rewritten	B4.2 To study boiling one digestive juice, which set should be the baseline? What must happen to the boiled liquid before comparison?	Inquiry design and experimental reasoning: choose the control for the question and control reaction temperature.	Keep the common content and skills, practised through self-study or static data.
Use “add 1 mL each of digestive juices X and Y” as the group for comparison. Boil only one digestive juice at a time. Complete both tests and record your observations.	Static data	B4.3 Use Figure 2 on A2 to record the heated sets and compare them with unboiled X + Y.	Observation: compare heating treatments with a baseline.	Keep the common content and skills, practised through self-study or static data.
Extension conclusion: the digestive juice that loses its function after boiling is_____.	Retained / equivalent rewrite	B4.4 Juice that loses its action after heating: choose the answer and explain why cooling does not restore its action.	Inference and content knowledge: explain heat treatment using enzyme denaturation.	Keep the common content and skills, practised through self-study or static data.
Do your observations support your first idea to test? Name the groups you compared and what you saw. Use the key learning points to improve your explanation.	Rewritten	B4.5 Use at least one main-set comparison and one heating comparison to explain emulsification (乳化作用), enzyme action and denaturation. Name the sets. Do not claim a measured combined effect or increase in speed.	Communication and content knowledge: explain concepts using named comparisons.	Concepts are taught first; assess their use with evidence rather than inventing an original hypothesis.

For teachers only. Do not give this part to students.

Worksheet feedback guide

Draft worksheet feedback guide. These marks are not equivalent to VL activity marks, SPSAI or CKT. Use separate common tests to assess research outcomes.

Item	Maximum	Shared skill / knowledge	Marking criteria
B1.1	2	Observation: separate what is seen from an explanation.	2: describes both layers and the boundary without claiming digestion; 1: one specific visible feature; 0: no relevant description.
B1.2	2	Inquiry design: judge whether a prediction is testable and whether its reason is sound.	2: separates testability from the wrong reason and explains enzyme action versus emulsification (乳化作用); 1: one useful judgement or explanation; 0: neither.
B1.3	4	Classification: identify the three kinds of variables.	Independent 1, dependent 1; five controlled factors at 0.4 each, total 2. A factor from a wrong category gives 0 for that category.
B1.4	1	Inquiry design and experimental reasoning: identify sound fixed conditions.	1 for all three correct conditions with no wrong choice; 0 for another attempted set.
B2.1	2	Inquiry design: explain the purpose of a control.	1 for the baseline, 1 for a sound reason to keep total volume fixed.
B2.2	2	Experimental reasoning and communication: read and label setups and volumes.	2 for all four correct setup/volume labels; 0.5 per complete label.
B2.3	3	Inquiry design and experimental reasoning: evaluate and correct an unfair procedure.	1 for the flaw, 1 for corrected volumes, 1 for the fairness explanation.
B3.1	4	Observation and communication: distinguish the four appearances.	1 per correctly described set, total 4.

B first-answer practice maximum: 36. Use C corrections for separate feedback; do not overwrite the first-answer marks.

Mark missing answers as unavailable and follow the study’ s missing-data rule. Do not invent evidence. Judge explanations and links to data, not keywords.

These marks are not equivalent to VL original-prediction, virtual-operation or completion marks and must not be converted to its 32-point score. Assess research outcomes using separate common tests.

For teachers only. Do not give this part to students.

Worksheet feedback guide

Draft worksheet feedback guide. These marks are not equivalent to VL activity marks, SPSAI or CKT. Use separate common tests to assess research outcomes.

Item	Maximum	Shared skill / knowledge	Marking criteria
B3.2	2	Inference: link an appearance with emulsification (乳化作用).	1 for the appearance; 1 for a sound explanation of emulsification (乳化作用) without molecular change.
B3.3	2	Inference: explain surface area without inventing speed data.	1 for increase, 1 for explaining the available surface and enzyme action.
B4.1	2	Inference and communication: distinguish clues, direct evidence and limits.	1 for the sound statement, 1 for a useful additional test or repeated evidence.
B4.2	2	Inquiry design and experimental reasoning: choose the control for the question and control reaction temperature.	1 for the unboiled baseline, 1 for cooling to the common temperature for a fair comparison.
B4.3	2	Observation: compare heating treatments with a baseline.	1 for each set.
B4.4	2	Inference and content knowledge: explain heat treatment using enzyme denaturation.	1 for Y, 1 for protein-enzyme denaturation and why cooling alone does not restore it.
B4.5	4	Communication and content knowledge: explain concepts using named comparisons.	1 each for emulsification (乳化作用), enzyme hydrolysis and denaturation; 1 for clear main/heating comparisons. Judge explanations, not keywords.

B first-answer practice maximum: 36. Use C corrections for separate feedback; do not overwrite the first-answer marks.

Mark missing answers as unavailable and follow the study’ s missing-data rule. Do not invent evidence. Judge explanations and links to data, not keywords.

These marks are not equivalent to VL original-prediction, virtual-operation or completion marks and must not be converted to its 32-point score. Assess research outcomes using separate common tests.